

1. A technology company offers its customers products and services.

Customer, **Product** and **Service** are entities/ relations because they are all subjects that the company will need to track information and interact with in specific ways.

2. Attributes of offering are offer ID and description.

Offering is an additional entity/relation because it has specified attributes: **ID** and **description**.

3. Attributes of products are product name, standard price and first release date.

The attributes of **Product** are clearly assigned: **product name**, **standard price** and **first release date**.

4. Attributes of services are the name of the company's unit responsible for the service and service conditions.

The company unit is the department that manages a certain area of the company/category of services.

The attributes of **Service** are assigned: **company unit**, **service condition**.

5. There are two different types of services: repair and maintenance.

- a. A repair service has a cost attribute and is the repair of some product.
- b. A maintenance service has an hourly rate attribute.

Here, Service is defined as a supertype and its subtypes are Repair and Maintenance.

The attributes of each subtype are clearly defined: **cost** and **hourly rate**.

6. When a customer purchases an offering, the company will need to keep track of the date that the offering was purchased and the contact person for that offering.

A relationship is established between **Customer** and **Offering**, in that a **Customer purchases** an **Offering**.

Additional attributes are given: **date of purchase**, **contact person**.

- These attributes could be for **Offering** or for the relationship **Purchase**. They will be attribute's of **Purchase** due to the relation not having any specified attributes.

7. **Attributes of customers are customer ID, name, DOB, age, address, phone numbers, and email.**

Attributes of customer are clearly assigned: [customer ID](#), [name](#), [DOB](#), [age](#), [address](#), [phone numbers](#), and [email](#).

8. **When a service is performed, that service is billed to some customer. Because some customers purchase offerings for their client, a customer may be billed for services he/she did not purchase, as well as for ones that were purchased.**

The relationship/associative entity [Purchase](#) will only relate [Customer](#) and [Offering](#). An additional relationship/associative entity is [Bill](#) which specifically relates [Service](#) to [Customer](#) and should identify the client if one is present.

[Client](#) serves as an attribute of [Bill](#).

9. **When a customer receives a bill for a service the company must keep track of the date the service was performed, the bill due date, and the amount due.**

Additional attributes of [Bill](#) are [date of service](#), [bill due date](#), and [amount due](#).

Customer is a relation with the PK, id, and attributes name, dob, age, address, email, and phone number.

Offering is a supertype relation with the attributes id and description. The description attribute serves as the subtype discriminator (aka, the input of the description determines which subtype the occurrence belongs to).

Product is a subtype of Offering. Its attributes are name, standard_price, and first_release_date. To connect Product to Offering, it inherits Offering's PK, ID, as a FK offering_id.

Service is a subtype of Offering. Its attributes are company_unit and service_condition. To connect Service to Offering, it inherits Offering's PK, ID, as a FK offering_id. Service is also a supertype. Its subtype discriminator is service_condition.

Maintenance is a subtype of Service. To connect Maintenance to Service, it inherits Service's inherited PK, offering_id, as a FK service_id. Its attribute is hourly_rate.

Repair is a subtype of Service. To connect Repair to Service, it inherits Service's inherited PK, offering_id, as a FK service_id. Its attribute is cost.

Purchase is an associative relation that connects Customer and Offering. Purchase shows what Offering a Customer buys/purchases. The attributes of Purchase are the date_of_purchase and contact_person for the purchase. Its PK is a composite of offering_id and customer_id.

Bill is an associative relation that connects Customer and Service. Bill shows the details of payment regarding a Service that Customer needs to know. The attributes of Bill are date_of_service, amount_due, and date_due. Because a Customer may purchase a Service on behalf client but still receive the Bill, this relation will track who the client for the service is; this attribute will be client_id. The PK of Bill will be a composite of customer_id and service_id.